ALICE L WALTON SCHOOL OF MEDICINE

Anemia of Chronic Disease/Inflammation

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Adapted from: Anemia of Chronic Disease/Inflammation

Learning Objectives (4) | Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of the mentioned anemia.
2. Explain the pathophysiology of the mentioned anemia.
3. Interpret clinical findings to properly diagnose the mentioned anemia.
4. Outline the management of the mentioned anemia.

CASE CONNECTION

TM is a 43-year-old female with rheumatoid arthritis (RA). She comes to the clinic to discuss her fatigue. "I'm tired all the time. When I get up in the morning, I dread the rest of the day because I'm so tired. On top of everything else, I feel like my arthritis has gotten worse." Physical exam shows mild conjunctival pallor and joint changes consistent with RA. Lab work is ordered, and the results show low hemoglobin and a normal mean corpuscular volume (MCV). The iron level and total iron-binding capacity are low, percent transferrin saturation is normal, and ferritin is high.

What is the most likely diagnosis? How would TM's condition be explained to her? Consider your answers as you read, and we'll revisit TM at the end of the brick.

GO TO CONCLUSION ↓

What Is Anemia of Chronic Disease/Inflammation?

Anemia of chronic disease (ACD), also called anemia of inflammation, is a decreased red blood cell (RBC) count as a result of an inflammatory chronic disease. This type of anemia can occur in many diseases, but the common thread is inflammation, which suppresses RBC formation and makes RBCs more susceptible to injury, shortening their lifespan.

ACD can arise from many conditions, including:

- Local and systemic infections (e.g., pneumonia, bacteremia)
- Cancer (e.g., leukemia, colon cancer)
- Autoimmune dis ure 1, systemic lupus erythematosus)

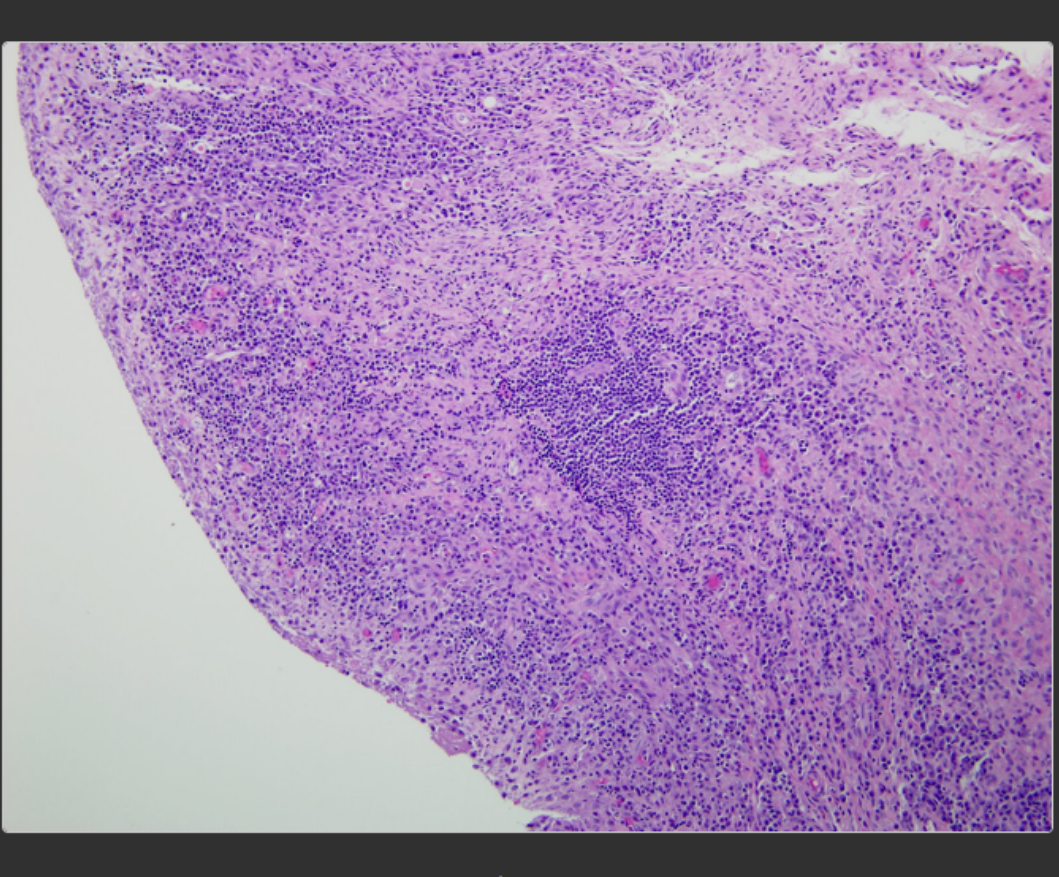


Figure 1

ACD is often part of a syndrome by which anemia occurs as a result of more than one mechanism. For example, inflammatory chronic gastrointestinal (GI) diseases such as Crohn disease and ulcerative colitis can cause anemia due to both chronic GI blood loss (resulting in iron deficiency) and inflammation.

ACD is the second most common type of anemia, after iron-deficiency anemia. The incidence increases with age, and older adults are more likely to have this type of anemia because they are also more likely to have one or more chronic diseases. About 1 mil of anemia.

The anemia is most often normocytic, but can be microcytic like iron-deficiency anemia.

What is the common process underlying most cases of anemia of chronic disease/inflammation?

How Do Patients with Anemia of Chronic Disease/Inflammation Present (LO #1)?

Patients with ACD may be asymptomatic, or they may present with common symptoms of anemia, including fatigue, weakness, dizziness, shortness of breath, headache, pallor, feeling cold, and/or tachycardia.

In addition to anemia symptoms, patients will also generally have one or more symptoms associated with the underlying chronic disease. For example, patients with rheumatoid arthritis may have joint pain and swelling, while those with cirrhosis i ng.

How would a patient with anemia in the context of rheumatoid arthritis present?

What Is the Pathophysiology of Anemia of Chronic Disease/Inflammation (LO #2)?

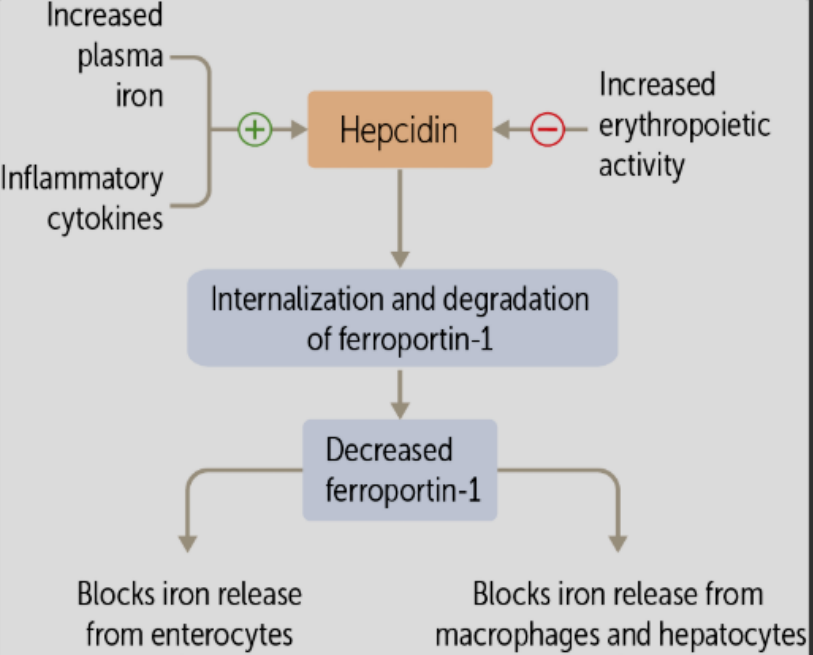
The pathogenesis of ACD is a bit more complex than other anemias. Chronic diseases can cause changes in RBCs that lead to both decreased erythrocyte production and a shortened RBC lifespan.

Inflammatory conditions associated with ACD include:

- Rheumatoid arthritis
- Inflammatory bowel disease (e.g., Crohn disease, ulcerative colitis)
- Systemic lupus
- Systemic sclerosis
- Chronic infectio
- Cancer (can also cause anemia via GI blood loss [e.g., colon cancer] or bone marrow infiltration [e.g., leukemias])

Decreased Erythrocyte Production

There are multiple reasons for reduced RBC production in ACD, but an important one is **reduced iron delivery to the bone marrow**. A large amount of the body's iron is stored in macrophages and liver cells. In chronic inflammation, there are high levels of interleukin 6 (IL-6), an inflammatory cytokine. IL-6 increases the production of **hepcidin** (Figure 2).


Increased plasma iron
Inflammatory cytokines
+
Hepcidin
−
Increased erythropoietic activity
Internalization and degradation of ferroportin-1
Decreased ferroportin-1
Blocks iron release from enterocytes
Blocks iron release from macrophages and hepatocytes
Decreased iron availability

Hepcidin regulates iron metabolism by:

- Preventing macrophages and liver cells from releasing stored iron
- Inhibiting iron transport across the basal membrane of intestinal cells, blocking iron transfer from the duodenal enterocyte into the plasma

As a result of these effects, a patient with chronic inflammation will have **decreased circulating iron** because iron is trapped in hepatocytes or macrophages or is not delivered from the intestine. Less iron in circulation means that less iron is delivered to the bone marrow, which means limited production of new RBCs (termed ineffective erythropoiesis).

In chronic inflammation, what prevents macrophages from releasing iron into the blood?

Decreased RBC Lif

In addition to reducing circulating iron, inflammation **reduces the RBC lifespan,** worsening the anemia. The reasons for this are not fully understood, but likely involve the action of inflammatory cytokines such as IL-1 and IL-6. These cytokines cause macrophages to prematurely ingest and destroy RBCs.

CLINICAL CORRELATION

The anemia of chronic kidney disease is usually considered a separate form of anemia from ACD but has some similarities. There is shortened RBC lifespan due to the uremic environment, as toxins build up in the blood due to poorly filtering kidneys. There is also ineffective erythropoiesis due to a lack of the hormone erythropoietin, which is made in the kidney. Less erythropoietin results in fewer RBCs produced. This form of anemia is treated with synthetic erythropoietin injections.

How Do We Diagnose Anemia of Chronic Disease/Inflammation (LO #3)?

Because ACD is secondary to an underlying disorder and does not have tests as specific as for other anemias, it is often a diagnosis of exclusion. ACD is diagnosed through laboratory tests, along with the history and physical exam, which may show signs and symptoms of the underlying disease. Testing for suspected inflammatory diseases with appropriate tests, such as the antinuclear antibody (ANA) test for lupus, may be performed. Nonspecific tests for inflammation include the erythrocyte sedimentation rate (ESR) and the C-reactive protein (CRP

Complete Blood Count

As in any anemia, the complete blood count (CBC) will show low blood hemoglobin, low hematocrit, and low RBC count. CBC indices will show a **normal or decreased mean corpuscular volume (MCV)**. A normal value is most common. A low MCV may lead to confusing ACD with iron-deficiency anemia, making the serum studies discussed in the next section important.

The peripheral blood smear typically does not show many abnormalities. RBCs may sometimes be hypochromic (pale) because they are deficient in hemoglobin.

Serum Iron Studies

The pattern of iron studies is helpful in making the diagnosis of ACD:

- **Serum ferritin** is normal or high because ferritin is a positive acute-phase reactant, meaning that more is made by the liver during inflammation. The high ferritin is helpful in distinguishing ACD from iron-deficiency anemia, which is characterized by a low serum ferritin.
- **Serum iron** is low because the iron is trapped in macrophages and hepatocytes. This is the same as in iron-deficiency anemia, so it is not helpful in distinguishing the two types.
- The **serum transferrin** (measured as the total iron binding capacity [TIBC]) is low since transferrin is a negative acute phase reactant, meaning that the liver makes less of it when inflammation is present.
- Because of the low circulating iron, the **transferrin saturation** percentage is most often low but can be normal since both transferrin and iron are reduced.

Table 1 summarizes these serum findings.

Table 1

Serum Study	Anemia of Chronic Disease
Iron	Low
Ferritin	Normal or high
TIBC	Low
Transferrin saturation percentage	Low (sometimes normal)

Which laboratory tests are used to help diagnose anemia of chronic disease/inflammation?

How Do We Manage Anemia of Chronic Disease/Inflam

The treatment of ACD is focused on addressing the underlying disorder. Iron-deficiency anemia should be ruled out using serum studies (e.g., serum ferritin), since some patients with chronic disease (such as those with colon cancer) may also be iron deficient.

Blood transfusions are generally not needed as long as the underlying cause can be treated.

How is anemia of chronic disease/inflammation generally treated?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to TM, what is her diagnosis? How would TM's condition be explained to her?

TM has anemia of c being rheumatoid arthritis (RA). TM's doctor would tell her that anemia is a condition in which there are not enough healthy red blood cells to carry oxygen throughout the body's tissues. This results in fatigue and weakness. TM asks if she should take an iron supplement and is told that the first step in treating this type of anemia is to treat the underlying disease. TM's doctor recommends a change in medication for the RA. TM feels hopeful that the new medication will allow the anemia to resolve, thereby improving her symptoms.

Summary

- Anemia of chronic disease/inflammation is a decreased red blood cell (RBC) count as a result of chronic inflammatory disease.
- Anemia of chronic disease (ACD) is the second most common type of anemia after iron-deficiency anemia.
- The anemia is most often normocytic, but can be microcytic (like iron-deficiency anemia).
- Patients may present with common anemia symptoms such as weakness, fatigue, dizziness, headache, shortness of breath, pallor, and feeling cold.
- In addition, patients will often have symptoms related to the underlying disease.
- Common causes of ACD include rheumatoid arthritis, systemic lupus, chronic infections, and malignancy.
- In ACD, the anemia is caused by reduced iron available to bone marrow; the iron is sequestered in the liver and macrophages due to increased hepcidin.
- Chronic inflammation may also shorten the RBC lifespan.
- When ACD is sus ammatory condition, a complete blood count (CBC) and iron studies are typically used for diagnosis.
- Iron indices show a low serum iron, high or normal ferritin, low total iron binding capacity (TIBC), and low transferrin saturation percentage.
- Treatment for ACD is focused on addressing the underlying disorder.

Review Questions

1. A patient with systemic lupus comes to the clinic with fatigue and mild shortness of breath when climbing stairs. Her complete blood count (CBC) shows a hemoglobin of 10.0 g/dL. Increased synthesis of which of the following most likely led to this anemia?

A. ○ Erythropoietin
B. ○ Haptoglobin
C. ● Hepcidin
D. ○ Hypoxia-inducible factor
E. ○ Protoporphyrin

Hide Explanation

The correct answer is hepcidin (C). This patient has a chronic inflammatory disease and anemia. In anem ine-mediated increase in hepcidin leads to a decrease in iron being released from macrophages and hepatocytes. This reduces circulating iron and prevents the bone marrow from using iron to make new red blood cells (RBCs). Hypoxia-inducible factor (D) stimulates RBC production via increased erythropoietin (A) in any form of anemia (except anemia of chronic kidney disease). However, this is an effect of anemia, not the cause of anemia of chronic disease (ACD). Haptoglobin (B) can bind to free hemoglobin after RBCs are hemolyzed intravascularly; its serum level drops in hemolytic anemias but is normal in ACD. Protoporphyrin (E) is a component of heme, but an increase is not responsible for ACD.

2. An increased value in which of the following tests is most likely in anemia of chronic disease/inflammation?

A. ● Ferritin
B. ○ Serum iron
C. ○ TIBC
D. ○ Transferrin saturation percentage

Hide Explanation

The correct answer is ferritin (A). In anemia of chronic disease (ACD), ferritin increases because it is an acute-phase reactant, with increased hepatic release due to inflammation. In ACD, the serum iron (B) and transferrin, measured as TIBC (C), are decreased due to transferrin saturation percentage (D) is dec sues.

3. Anemia of chronic disease (ACD) is caused primarily by a decrease in which of the following?

A. ○ Excretion of iron in the urine
B. ○ Oxidation of iron in hemoglobin
C. ● Release of iron from macrophages
D. ○ Storage of iron in the bone marrow
E. ○ Uptake of intestinal iron by enterocytes

Hide Explanation

The correct answer is release of iron from macrophages (C). In anemia of chronic disease (ACD), the release of iron from macrophages and hepatocytes is impaired because of increased hepcidin, resulting in less available iron for incorporation into red blood cell (RBC) precursors by the bone marrow. Excretion of iron in the urine (A) and oxidation of iron in hemoglobin (B) do not typically play a role in the pathogenesis of ACD. There is not increased storage of iron in the bone marrow (D) since the iron cannot reach it due to being trapped in hepatocytes and macrophages. In the gastrointestinal (GI) tract, uptake of intestinal iron by enterocytes (E) is normal, but it is unable to exit the cell.

4. A 26-year-old female presents as a follow-up to your clinic. She has a history of systemic lupus erythematosus (SLE), for which she takes multiple medications. Her CBC shows a hemoglobin of 10.0 g/dL (normal: 12.0-16.0 g/dL) with an MCV of 90 fL (normal: 80-100 fL). What treatment is required for her anemia?

A. ○ Oral iron supplementation
B. ○ Erythropoietin
C. ○ RBC transfusion
D. ● No treatment
E. ○ IV iron supplementation

Hide Explanation

The correct answer is no treatment (D). Oral iron supplementation (A) is only required if the patient has a documented iron deficiency, which we would need to confirm with iron studies. Erythropoietin (B) is generally not required for mild cases of anemia of chronic disease. Erythropoietin can be used in severe cases. RBC transfusion (C) is not indicated until the patient has a more profound anemia (typically not until patient is <7 g/dL). IV iron supplementation (E) is generally not required for mild cases of anemia of chronic disease. It can be used in severe cases.

How would you rate this brick?

References

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- Kumar V, Abbas AK, Aster JC. Hematopoietic and Lymphoid Systems. In: Kumar V, Abas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease.* 10th ed. Elsevier; 2021.
- Provan D, ed. *ABC of clinical haematology*. John Wiley & Sons, Incorporated; 2018.

Review Learning Objectives
Check off the objectives when you feel comfortable with the concept
01 ☑ ~~Differentiate~~ ~~mentioned anemia.~~
- Anemia of Chronic Disease - Symptoms, Causes, Treatment | NORD (rarediseases.org)
- Greer JP, et al, eds. *Wintrobe's Clinical Hematology.* 14th ed. Wolters Kluwer; 2018.
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04 ☑ ~~Outline the management of the mentioned anemia.~~

Objective Confidence: 4/4